

DAILY DRILLING REPORT SET

SYN-Assam-Duliajan-001 (W001) — SYN-Assam-Duliajan

1. REPORT SET IDENTIFICATION

Well ID	W001
Well Name	SYN-Assam-Duliajan-001
Field / Cluster	SYN-Assam-Duliajan
Area	Synthetic Assam Field Cluster (Duliajan-type operating area)
Latitude / Longitude	27.371219 N / 95.331752 E
Rig	SYN-RIG-E2000-01
Spud Date	09-Dec-2020
Final Report Date	20-Dec-2020
Daily Reports in Set	12 (Day 01 to Day 12)
Total Depth Reached	3,470 m
Primary Formation	Sandstone
Events Reported in Set	4

2. DEPTH PROGRESSION SUMMARY

Day	Date	From (m)	To (m)	Drilled (m)	Formation	Events
01	09-Dec-2020	0	345	345	Claystone	Cementing Issue
02	10-Dec-2020	345	805	460	Claystone / Shale	—
03	11-Dec-2020	805	1,215	410	Shale	—
04	12-Dec-2020	1,215	1,500	285	Shale	Stuck Pipe
05	13-Dec-2020	1,500	1,835	335	Shale	—
06	14-Dec-2020	1,835	2,145	310	Shale / Sandstone	—
07	15-Dec-2020	2,145	2,430	285	Sandstone	—
08	16-Dec-2020	2,430	2,695	265	Sandstone	—
09	17-Dec-2020	2,695	2,860	165	Sandstone	Mud Loss
10	18-Dec-2020	2,860	3,090	230	Sandstone	Stuck Pipe
11	19-Dec-2020	3,090	3,260	170	Limestone	—
12	20-Dec-2020	3,260	3,470	210	Limestone	—

PROTOTYPE TRAINING RECORD. Synthetic demonstration document prepared for the eRTMAC-NWIS prototype (SIH PS 26121). Depths, dates, parameters and events are generated from the prototype datasets and are not operational records of any company.

DOC No. DDR-W001-SET-R0 · PROTOTYPE TRAINING RECORD

DAILY DRILLING REPORT — DAY 01

W001 · 09-Dec-2020

Well ID: W001
 Well Name: SYN-Assam-Duliajan-001
 Report Date: 2020-12-09
 Day Number: 1
 Start Depth: 0 m
 End Depth: 345 m
 Drilled Interval: 345 m
 Formation: Claystone
 Mud Weight: 9.23 ppg
 ROP: 33.7 m/hr
 RPM: 145
 WOB: 6.6 klbf
 Events: Cementing Issue at 92 m (High)
 NPT: 18.3 hours

Operations performed.

- Rig up completed and well spudded at 09-Dec-2020. Drilled 17-1/2" hole from surface to 345 m.
- Slipped and cut drilling line; routine BOP function test carried out.
- Cementing Issue handled at 92 m — 18.3 hr non-productive time. Job evaluated against the pump/returns record; remedial squeeze / top-up cementing carried out and a cement bond log (CBL/VDL) run to confirm coverage.

Geological observations.

Cuttings throughout the interval are claystone, consistent with the 0–791 m unit logged in the offset wells of this cluster.

Mud / circulation observations.

Full returns maintained throughout the tour. Mud weight 9.23 ppg, properties held within programme; no losses or gains recorded.

Equipment / operational observations.

Casing/cementing equipment inspected after the event; cement unit and pressure recorders checked and found serviceable.

Events reported this day.

Event ID	Depth (m)	Formation	Type	Severity	Hr	Operational Impact
E0019	92	Claystone	Cementing Issue	High	18.3	Significant non-productive time; remedial operations and additional materials required.

End-of-day status.

Depth 345 m in Claystone. Drilling ahead. Non-productive time this tour 18.3 hr.

DAILY DRILLING REPORT — DAY 02

W001 · 10-Dec-2020

Well ID: W001
 Well Name: SYN-Assam-Duliajan-001
 Report Date: 2020-12-10
 Day Number: 2
 Start Depth: 345 m
 End Depth: 805 m
 Drilled Interval: 460 m
 Formation: Claystone / Shale
 Mud Weight: 9.80 ppg
 ROP: 30.3 m/hr
 RPM: 139
 WOB: 9.3 klbf
 Events: None
 NPT: 0.0 hours

Operations performed.

- Drilled 460 m of section from 345 m to 805 m (16.5 hr on bottom).
- Made connections and circulated bottoms-up at intervals; shakers monitored.
- Pulled out of hole, ran 13-3/8" casing to 790 m and cemented as programmed (Lead 12.6 ppg / Tail 15.8 ppg, cement to surface). Waited on cement and pressure tested.

Geological observations.

Formation change crossed during the tour: Claystone to 791 m, Shale to 1,906 m. Cuttings confirm the tops used for correlation with adjacent wells in the cluster.

Mud / circulation observations.

Full returns maintained throughout the tour. Mud weight 9.80 ppg, properties held within programme; no losses or gains recorded.

Equipment / operational observations.

All rig equipment functioning normally; no downtime recorded.

Events reported this day.

None. Operations proceeded within the planned parameter envelope.

End-of-day status.

DAILY DRILLING REPORT — DAY 03	W001 · 11-Dec-2020
Well ID: W001 Well Name: SYN-Assam-Duliajan-001 Report Date: 2020-12-11 Day Number: 3 Start Depth: 805 m End Depth: 1215 m Drilled Interval: 410 m Formation: Shale Mud Weight: 10.53 ppg ROP: 18.0 m/hr RPM: 130 WOB: 14.6 klbf Events: None NPT: 0.0 hours	
Operations performed. - Drilled 410 m of Shale from 805 m to 1,215 m (22.5 hr on bottom). - Made connections and circulated bottoms-up at intervals; shakers monitored.	
Geological observations. Cuttings throughout the interval are shale, consistent with the 791–1,906 m unit logged in the offset wells of this cluster.	
Mud / circulation observations. Full returns maintained throughout the tour. Mud weight 10.53 ppg, properties held within programme; no losses or gains recorded.	
Equipment / operational observations. Solids-control equipment cleaned; shaker screens changed. No downtime.	
Events reported this day. None. Operations proceeded within the planned parameter envelope.	
End-of-day status. Depth 1,215 m in Shale. Drilling ahead. No non-productive time this tour.	

DAILY DRILLING REPORT — DAY 04

W001 · 12-Dec-2020

Well ID: W001
Well Name: SYN-Assam-Duliajan-001
Report Date: 2020-12-12
Day Number: 4
Start Depth: 1215 m
End Depth: 1500 m
Drilled Interval: 285 m
Formation: Shale
Mud Weight: 11.04 ppg
ROP: 16.2 m/hr
RPM: 124
WOB: 17.0 klbf
Events: Stuck Pipe at 1370 m (Medium)
NPT: 12.8 hours

Operations performed.

• Drilled 285 m of Shale from 1,215 m to 1,500 m (9.7 hr on bottom).

• Made connections and circulated bottoms-up at intervals; shakers monitored.

• Stuck Pipe handled at 1,370 m — 12.8 hr non-productive time. String worked and jarred within the allowable overpull; lubricant pill spotted and hole cleaning improved with high-viscosity sweeps and increased rotation before back-reaming.

Geological observations.

Cuttings throughout the interval are shale, consistent with the 791–1,906 m unit logged in the offset wells of this cluster.

Mud / circulation observations.

Full returns maintained throughout the tour. Mud weight 11.04 ppg, properties held within programme; no losses or gains recorded.

Equipment / operational observations.

Overpull recorded on the string during the tour; jars worked and torque monitored. Top drive and drawworks otherwise functioning normally.

Events reported this day.

Event ID	Depth (m)	Formation	Type	Severity	Hr	Operational Impact
E0018	1,370	Shale	Stuck Pipe	Medium	12.8	Partial non-productive time; drilling suspended while the well was treated and conditioned.

End-of-day status.

Depth 1,500 m in Shale. Drilling ahead. Non-productive time this tour 12.8 hr.

Prototype Training Record · Synthetic demonstration data

Page 3 of 7

DAILY DRILLING REPORT — DAY 05

W001 · 13-Dec-2020

Well ID: W001
Well Name: SYN-Assam-Duliajan-001
Report Date: 2020-12-13
Day Number: 5
Start Depth: 1500 m
End Depth: 1835 m
Drilled Interval: 335 m
Formation: Shale
Mud Weight: 11.44 ppg
ROP: 15.4 m/hr
RPM: 122
WOB: 19.6 klb
Events: None
NPT: 0.0 hours

Operations performed.

- Drilled 335 m of Shale from 1,500 m to 1,835 m (22.5 hr on bottom).
- Circulated and conditioned mud; hi-vis sweep circulated bottoms-up.

Geological observations.

Cuttings throughout the interval are shale, consistent with the 791–1,906 m unit logged in the offset wells of this cluster.

Mud / circulation observations.

Full returns maintained throughout the tour. Mud weight 11.44 ppg, properties held within programme; no losses or gains recorded.

Equipment / operational observations.

BOP drill carried out with the crew; response time satisfactory.

Events reported this day.

None. Operations proceeded within the planned parameter envelope.

End-of-day status.

Depth 1,835 m in Shale. Drilling ahead. No non-productive time this tour.

DAILY DRILLING REPORT — DAY 06

W001 · 14-Dec-2020

Well ID: W001
Well Name: SYN-Assam-Duliajan-001
Report Date: 2020-12-14
Day Number: 6
Start Depth: 1835 m
End Depth: 2145 m
Drilled Interval: 310 m
Formation: Shale / Sandstone
Mud Weight: 11.76 ppg
ROP: 16.6 m/hr
RPM: 120
WOB: 21.3 klb
Events: None
NPT: 0.0 hours

Operations performed.

- Drilled 310 m of section from 1,835 m to 2,145 m (16.5 hr on bottom).
- Deviation survey taken; hole angle within tolerance.
- Pulled out of hole, ran 9-5/8" casing to 1,905 m and cemented as programmed (Lead 12.9 ppg / Tail 15.8 ppg, TOC 670 m). Waited on cement and pressure tested.

Geological observations.

Formation change crossed during the tour: Shale to 1,906 m, Sandstone to 3,090 m. Cuttings confirm the tops used for correlation with adjacent wells in the cluster.

Mud / circulation observations.

Full returns maintained throughout the tour. Mud weight 11.76 ppg, properties held within programme; no losses or gains recorded.

Equipment / operational observations.

All rig equipment functioning normally; no downtime recorded.

Events reported this day.

None. Operations proceeded within the planned parameter envelope.

End-of-day status.

Depth 2,145 m in Sandstone. 9-5/8" casing cemented at 1,905 m; waiting on cement.

DAILY DRILLING REPORT — DAY 07

W001 · 15-Dec-2020

Well ID: W001
Well Name: SYN-Assam-Duliajan-001
Report Date: 2020-12-15
Day Number: 7
Start Depth: 2145 m
End Depth: 2430 m
Drilled Interval: 285 m
Formation: Sandstone
Mud Weight: 12.18 ppg
ROP: 16.3 m/hr
RPM: 115
WOB: 23.3 klbf
Events: None
NPT: 0.0 hours

Operations performed.

- Drilled 285 m of Sandstone from 2,145 m to 2,430 m (22.5 hr on bottom).
- Deviation survey taken; hole angle within tolerance.

Geological observations.

Cuttings throughout the interval are sandstone, consistent with the 1,906–3,090 m unit logged in the offset wells of this cluster.

Mud / circulation observations.

Full returns maintained throughout the tour. Mud weight 12.18 ppg, properties held within programme; no losses or gains recorded.

Equipment / operational observations.

Solids-control equipment cleaned; shaker screens changed. No downtime.

Events reported this day.

None. Operations proceeded within the planned parameter envelope.

End-of-day status.

Depth 2,430 m in Sandstone. Drilling ahead. No non-productive time this tour.

DAILY DRILLING REPORT — DAY 08

W001 · 16-Dec-2020

Well ID: W001
Well Name: SYN-Assam-Duliajan-001
Report Date: 2020-12-16
Day Number: 8
Start Depth: 2430 m
End Depth: 2695 m
Drilled Interval: 265 m
Formation: Sandstone
Mud Weight: 12.52 ppg
ROP: 15.5 m/hr
RPM: 110
WOB: 25.5 klbf
Events: None
NPT: 0.0 hours

Operations performed.

- Drilled 265 m of Sandstone from 2,430 m to 2,695 m (22.5 hr on bottom).
- Slipped and cut drilling line; routine BOP function test carried out.

Geological observations.

Cuttings throughout the interval are sandstone, consistent with the 1,906–3,090 m unit logged in the offset wells of this cluster.

Mud / circulation observations.

Full returns maintained throughout the tour. Mud weight 12.52 ppg, properties held within programme; no losses or gains recorded.

Equipment / operational observations.

Mud pump No. 2 packing changed during routine circulation; no lost time.

Events reported this day.

None. Operations proceeded within the planned parameter envelope.

End-of-day status.

Depth 2,695 m in Sandstone. Drilling ahead. No non-productive time this tour.

DAILY DRILLING REPORT — DAY 09						W001 · 17-Dec-2020
Well ID: W001 Well Name: SYN-Assam-Duliajan-001 Report Date: 2020-12-17 Day Number: 9 Start Depth: 2695 m End Depth: 2860 m Drilled Interval: 165 m Formation: Sandstone Mud Weight: 12.78 ppg ROP: 15.6 m/hr RPM: 111 WOB: 26.9 klbf Events: Mud Loss at 2790 m (High) NPT: 11.9 hours						
Operations performed. - Drilled 165 m of Sandstone from 2,695 m to 2,860 m (10.6 hr on bottom). - Made connections and circulated bottoms-up at intervals; shakers monitored. - Mud Loss handled at 2,790 m — 11.9 hr non-productive time. Pumps slowed and a lost-circulation-material pill (fibrous/granular blend) spotted across the loss zone; mud weight trimmed to the minimum safe value and ECD reduced; pit levels monitored on a dedicated loss sheet.						
Geological observations. Cuttings throughout the interval are sandstone, consistent with the 1,906–3,090 m unit logged in the offset wells of this cluster.						
Mud / circulation observations. Returns reduced during the tour; losses treated with LCM and mud weight trimmed. Loss volumes recorded on the daily loss sheet. Mud weight held at 12.78 ppg after treatment.						
Equipment / operational observations. All rig equipment functioning normally; no downtime recorded.						
Events reported this day.						
Event ID	Depth (m)	Formation	Type	Severity	Hr	Operational Impact
E0001	2,790	Sandstone	Mud Loss	High	11.9	Significant non-productive time; remedial operations and additional materials required.
End-of-day status. Depth 2,860 m in Sandstone. Drilling ahead. Non-productive time this tour 11.9 hr.						

DAILY DRILLING REPORT — DAY 10						W001 · 18-Dec-2020
Well ID: W001 Well Name: SYN-Assam-Duliajan-001 Report Date: 2020-12-18 Day Number: 10 Start Depth: 2860 m End Depth: 3090 m Drilled Interval: 230 m Formation: Sandstone Mud Weight: 13.12 ppg ROP: 14.4 m/hr RPM: 106 WOB: 28.9 klbf Events: Stuck Pipe at 3061 m (Medium) NPT: 12.4 hours						
Operations performed. - Drilled 230 m of Sandstone from 2,860 m to 3,090 m (10.1 hr on bottom). - Circulated and conditioned mud; hi-vis sweep circulated bottoms-up. - Stuck Pipe handled at 3,061 m — 12.4 hr non-productive time. String worked and jarred within the allowable overpull; lubricant pill spotted and hole cleaning improved with high-viscosity sweeps and increased rotation before back-reaming.						
Geological observations. Cuttings throughout the interval are sandstone, consistent with the 1,906–3,090 m unit logged in the offset wells of this cluster.						
Mud / circulation observations. Full returns maintained throughout the tour. Mud weight 13.12 ppg, properties held within programme; no losses or gains recorded.						
Equipment / operational observations. Overpull recorded on the string during the tour; jars worked and torque monitored. Top drive and drawworks otherwise functioning normally.						
Events reported this day.						
Event ID	Depth (m)	Formation	Type	Severity	Hr	Operational Impact

DAILY DRILLING REPORT — DAY 11

W001 · 19-Dec-2020

Well ID: W001
Well Name: SYN-Assam-Duliajan-001
Report Date: 2020-12-19
Day Number: 11
Start Depth: 3090 m
End Depth: 3260 m
Drilled Interval: 170 m
Formation: Limestone
Mud Weight: 13.72 ppg
ROP: 7.4 m/hr
RPM: 92
WOB: 33.9 klb
Events: None
NPT: 0.0 hours

Operations performed.

- Drilled 170 m of Limestone from 3,090 m to 3,260 m (22.5 hr on bottom).
- Made connections and circulated bottoms-up at intervals; shakers monitored.

Geological observations.

Cuttings throughout the interval are limestone, consistent with the 3,090–3,470 m unit logged in the offset wells of this cluster.

Mud / circulation observations.

Full returns maintained throughout the tour. Mud weight 13.72 ppg, properties held within programme; no losses or gains recorded.

Equipment / operational observations.

BOP drill carried out with the crew; response time satisfactory.

Events reported this day.

None. Operations proceeded within the planned parameter envelope.

End-of-day status.

Depth 3,260 m in Limestone. Drilling ahead. No non-productive time this tour.

DAILY DRILLING REPORT — DAY 12

W001 · 20-Dec-2020

Well ID: W001
Well Name: SYN-Assam-Duliajan-001
Report Date: 2020-12-20
Day Number: 12
Start Depth: 3260 m
End Depth: 3470 m
Drilled Interval: 210 m
Formation: Limestone
Mud Weight: 14.03 ppg
ROP: 7.2 m/hr
RPM: 92
WOB: 35.8 klb
Events: None
NPT: 0.0 hours

Operations performed.

- Drilled 210 m of Limestone from 3,260 m to 3,470 m (16.5 hr on bottom).
- Made connections and circulated bottoms-up at intervals; shakers monitored.
- Pulled out of hole, ran 7" casing to 3,470 m and cemented as programmed (Tail 15.8 ppg, TOC 1,760 m). Waited on cement and pressure tested.
- Reached driller's total depth of 3,470 m. Circulated hole clean, logged and secured the well.

Geological observations.

Cuttings throughout the interval are limestone, consistent with the 3,090–3,470 m unit logged in the offset wells of this cluster.

Mud / circulation observations.

Full returns maintained throughout the tour. Mud weight 14.03 ppg, properties held within programme; no losses or gains recorded.

Equipment / operational observations.

Mud pump No. 2 packing changed during routine circulation; no lost time.

Events reported this day.

None. Operations proceeded within the planned parameter envelope.

End-of-day status.

Depth 3,470 m in Limestone. 7" casing cemented at 3,470 m; waiting on cement.